第九届中国计算机实践教育学术会议
暨第十八届全国高等学校计算机实践教学论坛
（CPEC2025）

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证书

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论文作者：樊光瑞 刘丹丹 张睿 潘理虎

通讯作者：樊光瑞

作者单位：太原科技大学

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国家级实验教学示范中心联席会计算机学科组
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RESEARCH

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The impact of AI-assisted pair programming on student motivation, programming anxiety, collaborative learning, and programming performance: a comparative study with traditional pair programming and individual approaches

Guangrui Fan[1*], Dandan Liu[2], Rui Zhang[1] and Lihu Pan[1]

Abstract

Purpose This study investigates the impact of AI-assisted pair programming on undergraduate students' intrinsic motivation, programming anxiety, and performance, relative to both human–human pair programming and individual programming approaches.

Methods A quasi-experimental design was conducted over two academic years (2023–2024) with 234 undergraduate students in a Java web application development course. Intact class sections were randomly assigned to AI-assisted pair programming (using GPT-3.5 Turbo in 2023 and Claude 3 Opus in 2024), human–human pair programming, or individual programming conditions. Data on intrinsic motivation, programming anxiety, collaborative perceptions, and programming performance were collected at three time points using validated instruments.

Results Compared to individual programming, AI-assisted pair programming significantly increased intrinsic motivation ($p < .001$, $d = 0.35$) and reduced programming anxiety ($p < .001$), producing outcomes comparable to human–human pair programming. AI-assisted groups also outperformed both individual and human–human groups in programming tasks ($p < .001$). However, human–human pair programming fostered the highest perceptions of collaboration and social presence, surpassing both AI-assisted and individual conditions ($p < .001$). Mediation analysis revealed that perceived usefulness of the AI assistant significantly mediated the relationship between the programming approach and student outcomes, highlighting the importance of positive perceptions in leveraging AI tools for educational benefits. No significant differences emerged between the two AI models employed, indicating that both GPT-3.5 Turbo and Claude 3 Opus provided similar benefits.

Conclusion While AI-assisted pair programming enhances motivation, reduces anxiety, and improves performance, it does not fully match the collaborative depth and social presence achieved through human–human pairing. These findings highlight the complementary strengths of AI and human interaction: AI support can bolster learning outcomes, yet human partners offer richer social engagement. As AI capabilities advance, educators should integrate

*Correspondence:
Guangrui Fan
fgr@tyust.edu.cn
Full list of author information is available at the end of the article

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RESEARCH

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Tool, tutor, or crutch?: A grounded theory of cognitive scaffolding and offloading in AI-assisted programming education

Dandan Liu[1,2*], Guangrui Fan[1*] and Lihu Pan[1*]

Abstract

Motivation The integration of generative AI into programming education has produced a widely reported tension between performance and learning. We distinguish immediate task performance from genuine learning: durable, transferable conceptual understanding and evaluative skill, and examine how AI support shapes learning processes, not merely outcomes. While many studies document improved speed, accuracy, and affect with AI support, questions remain about the quality of underlying learning.

Method and participants We use a constructivist grounded-theory design with constant comparison across two naturally occurring course sections (an AI-enabled section and a human pair-programming section used as a theoretical contrast). Over one semester, we collected triangulation across data sources from undergraduate Java programming students–interaction logs, pre-/post-course concept maps, and semi-structured dyadic interviews (AI-enabled: $N = 24$; theoretical contrast: $N = 17$).

Findings Analysis revealed a core tension between students' pursuit of "Domain Mastery" (conceptualization, explanation, and evaluation) and "Tool Mastery" (procedural efficiency with AI). We identified dynamic strategy switching (the Strategic Dance), Partnership Framing with an Illusion of Dialogue subtheme, and two recurrent evaluation challenges (Trust-but-Can't-Verify for novices; a Boilerplate Blindspot for more experienced students). We also describe attenuated meta-cognitive calibration–a mismatch between perceived readiness and independent capability–co-occurring with sustained offloading patterns. These categories synthesize into a process-level tension model with two recurrent loops (Scaffolding and Offloading), interpreted through Cognitive Load Theory and Self-Determination Theory.

Conclusion We offer a theory-building account that helps explain how widely observed performance and affect gains can co-occur with thinner opportunities for germane processing and authorship. The model generates testable implications (e.g., critique-the-AI phases, planned fading, verification journals), and we invite multi-site tests to evaluate boundary conditions.

*Correspondence:
Dandan Liu
S2134717@siswa.um.edu.my
Guangrui Fan
fgr@tyust.edu.cn
Lihu Pan
panlh@tyust.edu.cn

Full list of author information is available at the end of the article

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地址: [Fan, Guangrui; Zhang, Rui; Pan, Lihu] Taiyuan Univ Sci & Technol, Sch Comp Sci & Technol, 66 Waliu Rd, Taiyuan 030024, Shanxi, Peoples R China

[Liu, Dandan] Univ Malaya, Fac Arts & Social Sci, Dept Media & Commun Studies, Kuala Lumpur 50603, Wilayah Perseku, Malaysia

通讯作者地址: [Fan, Guangrui] (corresponding author), Taiyuan Univ Sci & Technol, Sch Comp Sci & Technol, 66 Waliu Rd, Taiyuan 030024, Shanxi, Peoples R China

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[Liu, Dandan] Univ Malaya, Fac Arts & Social Sci, Dept Media & Commun Studies, Kuala Lumpur 50603, Wilayah Perseku, Malaysia

通讯作者地址: [Fan, Guangrui] (corresponding author), Taiyuan Univ Sci & Technol, Sch Comp Sci & Technol, 66 Waliu Rd, Taiyuan 030024, Shanxi, Peoples R China

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